

Lifelogging in Mixed Reality

Adrian Aiordăchioae

MintViz Lab, MANSiD Research Center
Ștefan cel Mare University of Suceava
Suceava, Romania
adrian.aiordachioae2@student.usv.ro

Radu-Daniel Vatavu

MintViz Lab, MANSiD Research Center
Ștefan cel Mare University of Suceava
Suceava, Romania
radu.vatavu@usm.ro

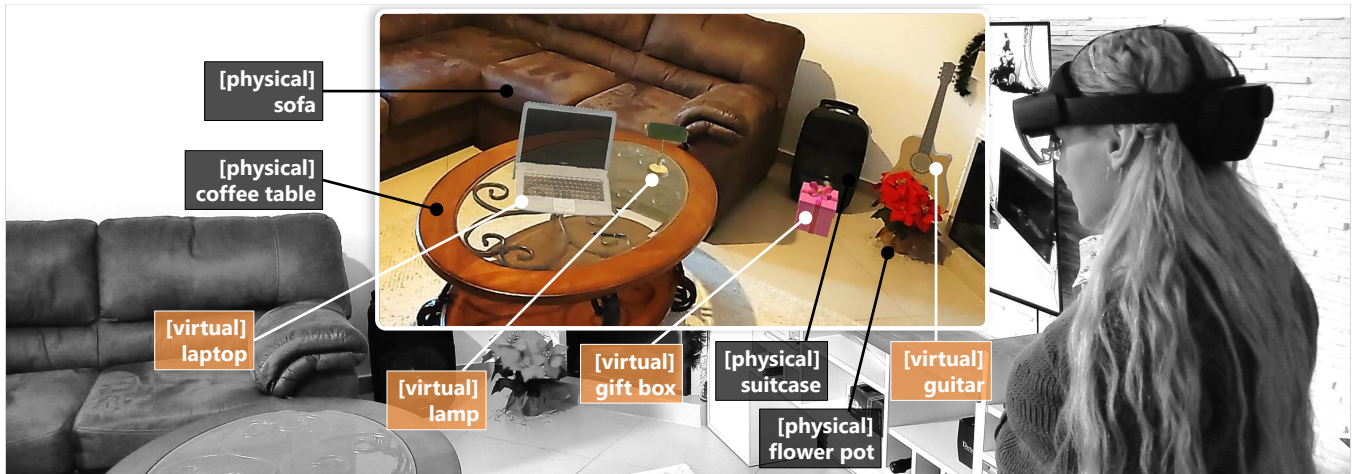


Figure 1: Mixed reality lifelogging aims to capture the experience of both the physical and virtual, e.g., in this figure, the perceptual experience of a physical coffee table, sofa, and flower pot, accompanied by a virtual laptop, guitar, and gift box. Note: the user's actual view of their mixed reality environment, as delivered through HoloLens v2, is shown in the close-up figure.

ABSTRACT

Lifelogging is a growing social phenomenon where personal devices capture records of one's life for future recalling and reminiscing. At the same time, mixed reality worlds are increasingly available due to advances in head-mounted display technology. However, no exploration has been conducted on lifelogging in such emerging worlds, where the lifelog should indiscriminately capture the experience of both the physical and virtual. In this paper, we introduce the concept of "mixed reality lifelogging" and specify it with the frameworks of the Metaverse Roadmap and broadcasting informational feeds as alternate realities. Second, we present a technical prototype for HoloLens. Third, we report findings from a preliminary experiment involving a lifelog recorded under varying densities of virtual objects anchored in the physical world. We conclude with future work opportunities for mixed reality lifelogging.

CCS CONCEPTS

• **Human-centered computing** → **Mixed / augmented reality;**
Mobile devices; Empirical studies in HCI.

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KEYWORDS

Lifelogging, mixed reality, augmented reality, head-mounted displays, smartglasses, HoloLens

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1 INTRODUCTION

Digital lifelogging [18] has become an increasing social phenomenon due to a growing diversity of personal digital devices—smart phones, smartglasses, action cameras, and others—coupled with cloud-based storage resources, which empower users to easily capture and archive records about their lives. The end result is personal multimedia archives, which indiscriminately represent lifeloggers' "every action, every event, every conversation, and every material expression" [15] with applications in providing memory assistance and augmentation [18,19,50] for recollecting, reminiscing, retrieving information, reflecting, and remembering intentions [45].

At the same time, see-through digital lenses integrated into smartglasses and head-mounted displays (HMDs) enable users to experience augmented [5] and mixed [33] reality worlds. Perception of such worlds involves distinctive aspects of collaboration, combination, and alignment of the physical and virtual [48] as well as switching between different mixtures of one's world view [38].

Also, by spanning many practical applications [8], mixed reality enables a new perspective on one's capabilities in the physical world, where context-dependent information is ubiquitously available [6].

At the intersection of these two emerging areas of personal computing, we focus on lifeloggers engaging in mixed reality, for which the lifelog should reflect perception of both the physical and virtual elements of the experienced world. For example, engaging in mixed reality gaming [49], interacting with virtual avatars [17], or watching augmented reality television [40], are life experiences with strong virtual world elements that should be reflected in the lifelog just like other experiences involving the physical world [18]. In this context, we define “mixed reality lifelogging” as the practice of recording one's actions, life events, and experiences in a mixed reality world rendered by a HMD or similar device; see Figure 1 for a visual illustration of the concept. Considering the growing availability of supporting technology,¹ these two areas of personal computing—centering on perception and recollection of one's life, respectively,—are likely to intersect in the foreseeable future.

In this context, we make several contributions: (1) we specify mixed reality lifelogging using two conceptual frameworks that integrate conventional lifelogging into broader contexts represented by Metaverse technology and specific forms of alternate realities; (2) we present the technical details of Lifelog-MR, our HoloLens application designed for mixed reality lifelogging; and (3) we report an evaluation of Lifelog-MR in the form of a preliminary experiment involving different conditions of densities of virtual objects placed in the physical world. Based on our results, we provide a discussion of future work opportunities for mixed reality lifelogging.

2 RELATED WORK

We relate to prior work on lifelogging and mixed reality, respectively, where the latter includes augmented reality [8], according to Milgram and Kishino's [33] Reality-Virtuality continuum.

Lifelogging, the practice of documenting various aspects of one's life using digital devices, has been described as a form of pervasive computing [15] due to its passive gathering and processing of life experience data [18], resulting in personal sociospatial archives that indiscriminately collect one's life and behavior [34]. Representative lifelogging applications include memory rehabilitation [50] and assistance [19], food journals [23], computer usage monitoring [20], assessment of life quality [57], and travel logs [4], to mention a few. We refer readers to Gurrin *et al.*'s [18] comprehensive survey of lifelogging as personal big data as well as surveys of lifelogging focused on applications in specific domains [19,25,42,50].

Mixed reality systems leverage context-dependent virtual content aligned with the physical environment towards augmented perception and amplified intelligence [5]. Representative applications span gaming [49], education [28], cultural heritage [32], entertainment [38], assistive technology [55], and psychological interventions [39]. For more details, we refer readers to Speicher *et al.*'s [48] examination of mixed reality concepts and Billinghurst *et al.*'s [8] comprehensive survey of augmented reality. In this space, relevant work to ours is represented by systems that implement video

queries in augmented and virtual reality. For example, Tesseract [30] provides a worlds-in-miniature interface for querying spatial design recordings, and “Who Put That There” [26] enables users to navigate through recorded time to view trajectories of objects that changed location. Other systems have centered on moment-reliving experiences by creating 3D scenes from 2D point-of-view videos, which can be visualized in augmented reality [27], or on authoring augmented experiences addressing specific audiences [41,52].

Despite a large scientific literature in each of these two areas, prior work has not explicitly focused on logging physical environments simultaneously with virtual content augmenting them for the purpose of lifelogging. In fact, none of the recent reviews on lifelogging [18,19,25,42,50] mention virtual, augmented, or mixed reality. A few works exist on visualizing lifelogs in VR/AR [22,43,44], but they do not address life recording in such environments. However, when the lifelogger's experience with their environment does involve virtual content, the lifelog should accordingly reflect this experience as an “undiscriminating collection of information concerning one's life and behaviour” [34] that documents “every moment” of users' lives [42]. The focus here is on the word “every,” which encompasses life actions, events, and experiences of virtual content alike when the lifelogger's world intertwines the physical and virtual in a unified perceptual experience. Thus, mixed reality lifelogging capitalizes on prior results from both the areas of lifelogging and mixed reality, putting together a pipeline of techniques explored so far in isolation—mixed reality capture, analysis, querying of recordings, and so forth—to support the social phenomenon of lifelogging when users engage in new, emerging worlds.

In this context, two generic frameworks are useful to connect to this novel concept. One is the Metaverse Roadmap (MVR) [47], introduced to structure the technology of computer-supported social spaces. MVR specifies two technological continua—intimate vs. external and simulation vs. augmentation—in a taxonomy of application domains relating to the Metaverse, where lifelogging appears at the intersection of intimate and augmentation technology since “augmentation technologies record and report the intimate states and life histories of objects and users” [47]. A second framework useful to our scope is the Alternate Reality Broadcast-Time (ARTB) [1] conceptual space, where lifelogging is described as asynchronous informational feed addressing remote audiences to enable the experience of a multimodal alternate reality [11]. For lifelogging, “remote” means asynchronous in the time domain, while the audience is composed, at the minimum, of the lifeloggers themselves. Although not addressing mixed reality lifelogging explicitly, these two frameworks approach the closest our scope and enable us to relate life recording to computer-supported worlds; see next.

3 LIFELOG-MR

We introduce Lifelog-MR, our system designed for lifelogging in mixed reality. We start by contextualizing mixed reality lifelogging, and continue with presenting the technical details of Lifelog-MR.

3.1 What Does Lifelogging Entail in Mixed Reality?

Lifelogging in mixed reality environments positions at the intersection of (i) experiencing the world as a mixture of both physical

¹The global action camera market was valued at 2.2 billion USD in 2022 (<https://www.statista.com/statistics/1059920/worldwide-sales-action-cams>), while the AR headset and glasses consumer market was valued at 2.4 billion USD in 2023 (<https://www.statista.com/topics/10134/ar-glasses>).

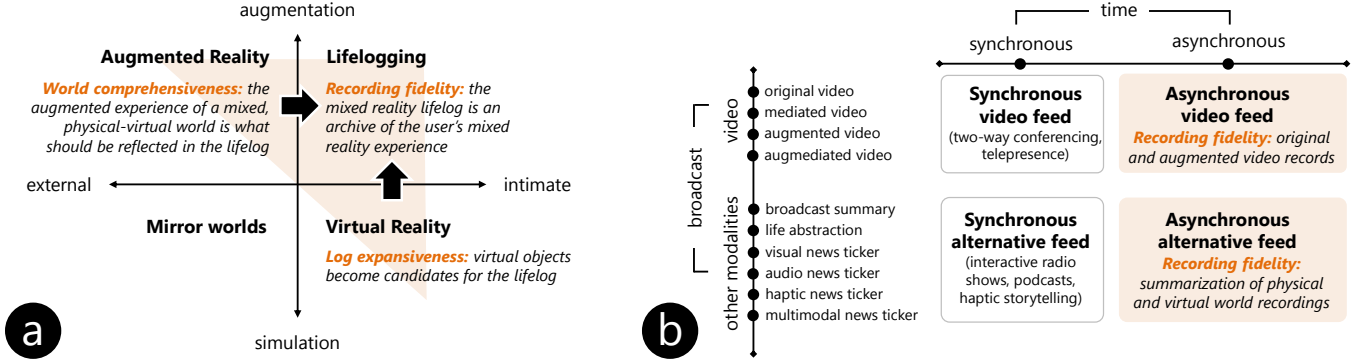


Figure 2: Mixed reality lifelogging within the taxonomy of the Metaverse Roadmap [47] (a) and the Alternative Reality Broadcast-Time conceptual space [1] (b)—two frameworks that integrate generic lifelogging into broader contextual domains.

and virtual elements [33] and (ii) the phenomenon of lifelogging, centered on capturing and archiving records about one's life experiences [18]. In the following, we connect to the two conceptual frameworks, MVR [47] and ARBT [1], presented in Section 2, to characterize lifelogging systems in mixed reality environments.

In the MVR [47] space, lifelogging is characterized by *intimate* technology, designed to address the identity of the individual through personal records, as well as *augmentation* technology, designed to expand the perception of one's environment through new sensing capabilities; see the top-right quadrant in Figure 2a. In this space, mixed reality lifelogging generates innovation towards “delivering the entire metaverse,” according to Billingham's [7] view on MVR, by spanning three quadrants: virtual reality, augmented reality, and conventional lifelogging, as follows. Virtual objects, native of a virtual world (bottom-right quadrant in Figure 2a), are anchored in the physical world to enable augmented perception (top-left quadrant), an experience that should be indiscriminately captured in the lifelog (top-right quadrant). In the ARBT [1] space, lifelogging is an informational feed characterized by its asynchronicity with respect to the moment of recording; see the two left quadrants in Figure 2b. The records, whether video or using alternative modalities, must ensure recording fidelity by encompassing an indiscriminate representation of both the physical and virtual.

Based on these two theoretical frameworks, we identify three qualitative characteristics of mixed reality lifelogging: *log expansiveness* (i.e., virtual objects become candidates for the lifelog), *world comprehensiveness* (i.e., the mixed reality lifelog corresponds to a mixed reality world that comprehensively combines both physical and virtual elements) and, as a consequence of the former, *recording fidelity* of the mixed reality life experience that is captured in the lifelog. Next, we present the technical implementation details of Lifelog-MR, our system for mixed reality lifelogging.

3.2 Technical Implementation of Lifelog-MR

Based on our analysis of conventional lifelogging systems [2,3,13,18,24] and the specifics of our application domain, discussed in Subsection 3.1, we identified the following design requirements for Lifelog-MR: (DR₁) always on, passive and automatic visual capture [29] of one's mixed reality [33], (DR₂) recording from the first

person, eye-level perspective [18], (DR₃) privacy orientation by easily switching recording on and off [13,24], and (DR₄) synchronization of data collected from the physical and virtual worlds for a unified perceptual experience of the mixed reality informational feed [1]. We developed Lifelog-MR using HoloLens v2,² a HMD that integrates many features convenient to our application domain, e.g., see-through lenses with high resolution, spatial mapping for stable and persistent anchoring of virtual objects in the physical world, video recording at 2K@30fps, Wi-Fi connectivity, and battery life supporting between two and three hours of active use. Lifelog-MR incorporates several software components and workflows.

3.2.1 Mixed reality set up. This component enables setting up the mixed reality environment by anchoring virtual objects onto physical surfaces. The workflow consists of the user navigating to a specific location in the physical world, summoning a virtual object with a voice command, e.g., “airplane,” and subsequently using standard HoloLens mid-air gesture input to resize and position the virtual object in the physical world; see Figure 3a for an example.

3.2.2 Lifelog recording and analysis. Two types of data are recorded in Lifelog-MR: (i) eye-level perspective video of the physical environment using the HoloLens built-in video camera (following design requirements DR₁ and DR₂) and (ii) virtual objects falling within the user's field of view (DR₁ and DR₄); see Figure 3b for an example of the user's view of the mixed reality environment. Subsequently, individual snapshots are extracted from the video of the physical world at 2 fps, according to tag-based lifelog recording recommendations from [2], and are submitted to a computer vision service [14] to automatically identify physical-world objects and concepts, e.g., “door” and “hallway” for the example in Figure 3b. Virtual objects, e.g., “sofa” and “guitar,” are also registered at the same rate of 2 fps with corresponding timestamps to ensure synchronization with data recorded from the physical world (DR₄). These records are stored in the Firebase Realtime Database and Firebase Storage. Recording is controlled using the “start” and “stop” voice commands (DR₃). While our implementation targeted HoloLens, which is a see-through HMD, the design requirements can be satisfied with other devices, including pass-through HMDs.

²<https://www.microsoft.com/en-us/hololens>



Figure 3: Component workflows of Lifelog-MR: (a) setting up the mixed reality environment by anchoring virtual objects, e.g., the airplane, via gesture and voice input; (b) HoloLens-mediated view of the world during lifelog recording, e.g., a physical hallway with a virtual sofa and guitar; and (c) lifelog data summarization as a word cloud presented in a smartphone application.

3.2.3 Lifelog query. We developed a mobile application for Android that enables running queries in the mixed reality lifelog based on keywords, e.g., “sofa,” and time intervals of the recording, e.g., records between October and December 2023, which are summarized as word clouds; see Figure 3c. Word clouds, where concepts are displayed according to their appearance frequency in the lifelog, enable lifelog summarization [18] and act as tags or links [2] to actual video recordings from which they were extracted.

4 PRELIMINARY EXPERIMENT

We conducted a preliminary experiment to evaluate our prototype for mixed reality lifelogging as well as to gain insights into future opportunities for engineering interactive systems in this space, on which we elaborate in Section 5. We designed the technical evaluation of our prototype following methodology that has been previously used for evaluating lifelogging systems [2,3].

4.1 Participant

We recruited one volunteer (male, 33 years old), referred to in the following with the fictive name Christopher, to use Lifelog-MR in an indoor, office building scenario.³ Noteworthy, one participant was sufficient for the purpose of our preliminary experiment to collect the necessary data since our need for a human subject was simply to wear the HMD. Also, studies with one participant only are not uncommon for evaluating lifelogging systems, e.g., Aiordăchioae and Vatavu [2,3] used one participant to evaluate LifeTags, Zhou *et al.* [56] evaluated the LIFER interactive lifelog retrieval system with one user, and Microsoft’s SenseCam lifelogging device [21] was originally evaluated with just one patient with amnesia.

4.2 Design

Our evaluation took the form of a repeated-measures design with one independent variable, MR-COMPLEXITY, ordinal with three conditions—*low*, *medium*, and *high*,—representing mixed reality environments with increasingly more virtual objects. For the specific case of the three-floor office building where we conducted the evaluation, these conditions were assigned to each of the three floors,

as follows: 14 virtual objects were distributed on the ground floor, corresponding to an average density of about 2 virtual objects per 100m² in the *low* complexity condition, 28 virtual objects were distributed inside a similar area on the first floor in the *medium* condition, and 56 objects on the second floor in the *high* condition, following a geometric progression with common ratio 2. This distribution resulted in the *high* complexity condition being a fourfold more densely populated environment than the *low* complexity one. Our dependent variable was NUM-RECORDS, representing the number of objects and concepts extracted from the video recording of the physical world and, respectively, the number of virtual objects located within the user’s field of view at any specific timestamp. For example, NUM-RECORDS is 7 for the top-left photograph in Figure 4, where five physical objects and concepts (“palm tree,” “architecture,” “windows,” “door,” and “furniture”) and two virtual objects (the two birds on the physical palm tree branches) are present.

4.3 Procedure

We compiled a set of 98 virtual objects of various types—furniture and appliances, but also animals and human avatars—from specialized platforms [16,46]. The virtual objects were randomly assigned to each MR-COMPLEXITY condition and distributed on each of the three floors; see Figure 4 for photographs taken during the evaluation. Christopher’s role was to walk along the hallways for 30 minutes with 10 minutes per floor. During this time, our application recorded 2K@30fps videos of the physical world from the eye-level perspective and registered, with timestamps, the virtual objects located within Christopher’s field of view at 2 fps. After the recording, we processed the videos of the physical world using a custom Python application to extract individual frames at a rate of 2 fps, according to recommendations from [2], and submitted them to a machine vision service [14] to automatically extract objects and concepts. This procedure led to two logs, corresponding to the physical and virtual worlds, respectively, which we synchronized based on the recorded timestamps to create the mixed reality lifelog.

4.4 Results

We report results from a dataset consisting of 3,600 images = 3 (MR-COMPLEXITY conditions) × 10 (minutes of recording per condition)

³Outdoor lighting causes instability in the HoloLens built-in tracker; see <https://learn.microsoft.com/en-us/hololens/hololens-environment-considerations>.



Figure 4: Snapshots captured during the evaluation of Lifelog-MR, corresponding to low (a), medium (b), and high (c) complexity mixed reality environments according to the density of virtual objects anchored in the physical world.

× 60 (seconds) × 2 (fps), extracted from the video recordings. The extraction process resulted in a total number of 35,185 records representing objects and concepts from the physical world detected with a confidence level above 95% [14], of which 184 were unique. We also logged a total number of 15,270 records of virtual objects identified within Christopher’s field of view, which encompassed the totality of our set of 98 objects (see Subsection 4.3). Figures 5a and 5c show the number of records stored in the virtual and physical parts of the lifelog at specific timestamps, which confirm continuous operation of our application and Christopher’s active walking and exploration of the office building environment.

As Christopher traversed the hallways of the three floors, different areas exhibiting varying densities of virtual objects are clearly identifiable in Figure 5a and, as a summary, in Figure 5b. At the same time, the number of records extracted from the physical reality remained relatively constant across all of the three experimental conditions; see Figures 5c and 5d. In total, we recorded 1,853 virtual objects in the *low* complexity condition, 2.7 more objects in the

medium condition, and 4.5 more objects in the *high* complexity condition. These results reflect our specific design of the experimental conditions following a geometric progression with common ratio 2 and, consequently, validate the lifelog recording.

Figure 6 shows word clouds compiled from the data extracted from the physical and virtual worlds as well as their union in the mixed reality lifelog. The word cloud corresponding to the physical world (Figure 6a) reveals objects and concepts representative of an indoor office building environment, such as “architecture,” “indoors,” “room,” “furniture,” “window,” “interior,” “contemporary,” and so forth and, thus, qualitatively validates the recording of Christopher’s physical environment. The word cloud corresponding to the virtual world (Figure 6b) comprises virtual objects from our set, displayed in text sizes corresponding to their appearance frequency within Christopher’s field of view. Finally, Figure 6c shows the union of the records from the physical and virtual word clouds, corresponding to a data summarization [18] of the mixed reality lifelog. This last word cloud reveals what Christopher’s unified perceptual experience of both the physical and virtual was like, where

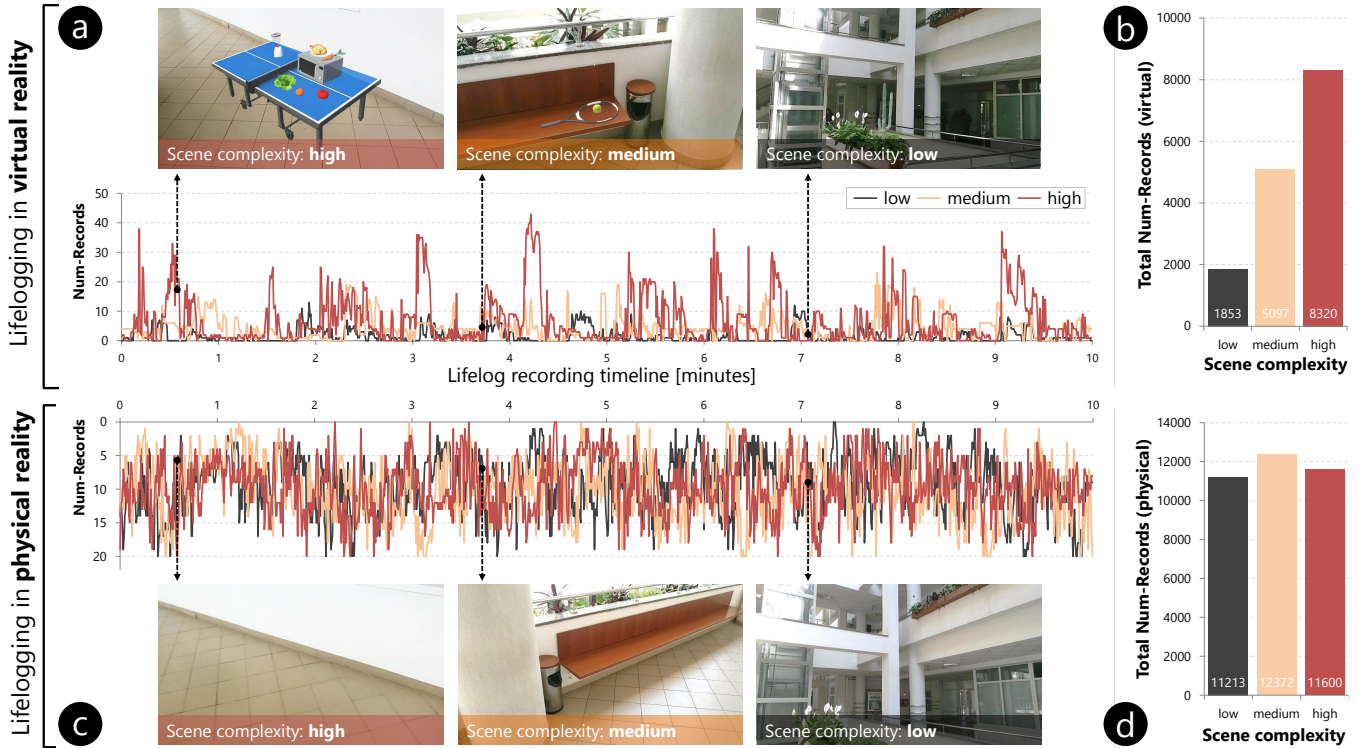


Figure 5: Lifelog recording timeline corresponding to the virtual (a) and physical (c) worlds and total number of records (b,d).



Figure 6: Word clouds generated in our preliminary experiment from $N=15,270$ and $N=35,185$ records, respectively, corresponding to the physical (a) and virtual (b) worlds, and word cloud from $N=50,455$ records summarizing the mixed reality lifelog (c).

virtual objects, such as “chicken,” “guitar,” or “spider,” were intermixed with the physical environment represented by the “modern,” “contemporary” “architecture” of an “indoor” space.

5 DISCUSSION

Our results showcase the viability of mixed reality lifelogging, readily achievable using current HMD technology. Furthermore, they highlight the potential for wearable computer systems at the intersection of two emerging areas of personal computing—lifelogging, focused on self-recording, and mixed reality, aimed at augmenting the environmental perception in relation to the self. Despite their strong connection around the *self*, these two areas of computing

have remained unexplored in conjunction. Hence, we seize this opportunity to complement our theoretical exploration and empirical results with a discussion about the technical limitations of our prototype and suggestions for future developments in this space.

5.1 Limitations

Our preliminary experiment for the technical evaluation of Lifelog-MR was conducted with a limited set of virtual objects of limited nature. However, other types of virtual content relevant for mixed reality lifelogging, such as gaming [49], virtual avatars [17], or virtual TV screens [40] displayed in the physical world, represent interesting future explorations. Also, our prototype considered only life

tags [2] to represent the lifelog, a format suitable in our evaluation due to its simplicity and capacity to capture the essence of the experienced environment, but alternative visualizations may be more suited for querying and consuming mixed reality lifelog content, such as worlds-in-miniature approaches [30], visualizations of proxemic dimensions [9], or object-based navigation techniques [26]. Finally, our evaluation experiment was limited in scope and duration. Whereas mixed reality lifelogging encompasses the totality of one's actions, life events, and experiences, our evaluation primarily focused on recording virtual and physical objects, reducing actions, events, and experiences to mere concepts. Larger lifelogs, resulting from users engaging for longer periods of time in mixed reality environments, are also recommended in future evaluations of mixed reality lifelogging. Next, we present more future work opportunities, structured according to the technological, experiential, and conceptual dimensions of mixed reality lifelogging.

5.2 Future Work Opportunities

At the *technological level*, we see interesting opportunities for collecting richer lifelogs by leveraging more sophisticated HMD designs. These include incorporating additional sensors into HMDs, such as multiple video cameras [53] to record multi-perspective video [3], leveraging HMD networks for capturing the visual perspective of virtual avatars, such as during holographic teleportation [35], and the capability of running simulations of lifelogging in extended reality environments through dedicated platforms [36].

At the *experiential level*, conducting extended user studies of mixed reality lifelogging will uncover users' needs, preferences, and expectations for such systems, according to various demographics, but also enable examination of how queries operate within the rich informational space of mixed reality lifelogs. Furthermore, the impact of the nature of the recorded content, physical or virtual, in the mixed reality lifelog on the user experience described in terms of the five R's [45]—recollecting, reminiscing, retrieving information, reflecting, and remembering intentions—is interesting to examine in close detail: are records of physical and virtual perceived separately or do they form a unified experience of past life events?

At a *conceptual level*, we see intriguing extensions of our concept towards “mediated reality lifelogging,” wherein the recorded lifelog does not replicate the actual physical reality, but instead the user's mediated perception of it [31]. Future work in this direction encompasses archiving records of diminished reality [12], exploring the intersection between augmented reality and ambient intelligence [51], ability-based enhancements of visual reality [37], kinesthetic feedback as a form of extended reality delivered through body-augmentation wearable devices [10], and visual cues [54] meant to guide perception of physical reality—which are key to the lifelogger as drivers of their perception at the time of recording.

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